6.6: The cytoskeleton is a network of fibers that organizes structures and activities in the cell

In the early days of electron microscopy, biologists thought that the organelles of a eukaryotic cell floated freely in the cytosol. But improvements in both light microscopy and electron microscopy have revealed the **cytoskeleton**, a network of fibers extending throughout the cytoplasm **(Figure 6.20)**. Bacterial cells also have fibers that form a type of cytoskeleton, constructed of proteins similar to eukaryotic ones, but here we will concentrate on eukaryotes. The eukaryotic cytoskeleton plays a major role in organizing the structures and activities of the cell.

Figure 6.20

```
The cytoskeleton.
```

Microtubules appear as a network of string-like structures concentrated around the nucleus (diameter: about 8 micrometers) and microfilaments are clustered toward the perimeter of the cell.

As shown in this fluorescence micrograph, the cytoskeleton extends throughout the cell. The cytoskeletal elements have been tagged with different fluorescent molecules: green for microtubules and reddish-orange for microfilaments. A third component of the cytoskeleton, intermediate filaments, is not evident here. (The blue color tags the DNA in the nucleus.)

BioFlix® Animation

Cytoskeleton

image

Play

image

Roles of the Cytoskeleton: Support and Motility

The most obvious function of the cytoskeleton is to give mechanical support to the cell and maintain its shape. This is especially important for animal cells, which lack walls. The remarkable strength and resilience of the cytoskeleton as a whole are based on its architecture. Like a dome tent, the cytoskeleton is stabilized by a balance between opposing forces exerted by its elements. And just as the skeleton of an animal helps fix the positions of other body parts, the cytoskeleton provides anchorage for many organelles and even cytosolic enzyme molecules. The cytoskeleton is more dynamic than an animal skeleton, however. It can be quickly dismantled in one part of the cell and reassembled in a new location, changing the shape of the cell.

Some types of cell motility (movement) also involve the cytoskeleton. The term *cell motility* includes both changes in cell location and movements of cell parts. Cell motility generally requires interaction of the cytoskeleton with **motor proteins**. There are many such examples: Cytoskeletal elements and motor proteins work together with plasma membrane molecules to allow whole cells to move along fibers outside the cell. Inside the cell, vesicles and other organelles often use motor protein “feet” to “walk” to their destinations along a track provided by the cytoskeleton. For example, this is how vesicles containing neurotransmitter molecules migrate to the tips of axons, the long extensions of nerve cells that release these molecules as chemical signals to adjacent nerve cells **(Figure 6.21)**. The cytoskeleton also manipulates the plasma membrane, bending it inward to form food vacuoles or other phagocytic vesicles.

Figure 6.21

Motor proteins and the cytoskeleton.

Motor proteins and the cytoskeleton.

Part a. Vesicle attached to motor protein at a receptor point. Arrows from an A T P indicate that a motor protein steps along microtubule using energy from A T P. Microtubule drawn as hollow tube.

Part b. Two vesicles containing neurotransmitters move along microtubule toward tip of a nerve cell extension called an axon (SEM). Microtubule visible as cable-like structure. Vesicles are globular structures atop the microtubule. Vesicles are globular structures approximately 0.25 micrometers long atop the microtubule.

Interview

Interview with George Langford: Investigating how motor proteins move cargo

George Langford.

Components of the Cytoskeleton

Now let's look more closely at the three main types of fibers that make up the cytoskeleton: *Microtubules* are the thickest of the three types; *microfilaments* (also called actin filaments) are the thinnest; and *intermediate filaments* are fibers with diameters in a middle range **(Table 6.1)**.

Table 6.1

The Structure and Function of the Cytoskeleton

A table with three fluorescent micrographs of fibroblasts are given.

Property	Microtubules (Tubulin polymers)	Microfilaments (actin filaments)	Intermediate filaments
Structure	Hollow tubes	Two intertwined strands of actin	Fibrous proteins coiled into cables
Diameter	25 nanometers with 15 nanometer lumen	7 nanometers	8 to 12 nanometers
Protein subunits	Tubulin, a dimer consisting of an alpha tubulin and a beta tubulin	Actin	One of several different proteins (including keratins)
Main functions	Maintenance of cell shape; cell motility; chromosome movements in cell division; organelle movements	Maintenance of cell shape; changes in cell shape; muscle contraction; cytoplasmic streaming (plant cells); cell motility; cell division (animal cells)	Maintenance of cell shape; anchorage of nucleus and certain other organelles; formation of nuclear lamina
Fluorescence micrographs of fibroblasts. Connective tissue cells called fibroblasts are a favorite cell type for cell biology studies because they spread out flat and their internal structures are easy to see. In each	Image shows cluster of 7 fibroblasts (diameter of each cell: about 10 micrometers) filled with networks of microtubules. Below image is a diagram of a hollow tube representing a microtubule and a cutaway drawing zooming	Actin filaments networked across a fibroblast (diameter about 10 micrometers). Diagram zooms in on filament shown as a coil of two thick strands. Zooming in on the structure reveals a coil of spherical actin	Image of fibroblast (about 10 micrometers in diameter) with intermediate filaments and nucleus visible (about 5 micrometers in diameter). Cutaway diagrams zoom in on intermediate filament structure. Intermediate

fibroblast shown here, the structure of interest has been tagged with fluorescent molecules. The D N A in the nucleus has also been tagged in the first micrograph (blue) and third micrograph (orange).	in on microtubule structure as a coil of columns of tubulin dimers. Tubulin dimers are made of two subunits: alpha and beta tubulin. Diameter of the microtubule = 25 nanometers.	subunits pushed together like beads (diameter 7 nanometers).	filament structure shown as a thick tube made up of keratin proteins with fibrous subunits (keratins coiled together). Filament diameter = 8-12 nanometers.

VISUAL SKILLS *How many tubulin dimers are in the boxed row?*

*For suggested answer, see*Appendix A.**

Microtubules

All eukaryotic cells have **microtubules**, hollow rods constructed from globular proteins called tubulins. Each tubulin protein is a *dimer*, a molecule made up of two components. A tubulin dimer consists of two slightly different polypeptides, α-tubulin and β-tubulin. Microtubules grow in length by adding tubulin dimers; they can also be disassembled and their tubulins used to build microtubules elsewhere in the cell. Because of the orientation of tubulin dimers, the two ends of a microtubule are slightly different. One end can accumulate or release tubulin dimers at a much higher rate than the other, thus growing and shrinking significantly during cellular activities. (This is called the “plus end,” not because it can only add tubulin proteins but because it’s the end where both “on” and “off” rates are much higher.)

Microtubules shape and support the cell and also serve as tracks along which organelles equipped with motor proteins can move. In addition to the example in Figure 6.21, microtubules guide vesicles from the ER to the Golgi apparatus and from the Golgi to the plasma membrane. Microtubules are also involved in the separation of chromosomes during cell division, as shown in Figure 12.7.

Centrosomes and Centrioles

In animal cells, microtubules grow out from a **centrosome**, a region that is often located near the nucleus. These microtubules function as compression-resisting girders of the cytoskeleton. Within the centrosome is a pair of **centrioles**, each composed of nine sets of triplet microtubules arranged in a ring **(Figure 6.22)**. Although centrosomes with centrioles may help organize microtubule assembly in animal cells, many other eukaryotic cells lack centrosomes with centrioles and instead organize microtubules by other means.

Figure 6.22

Centrosome containing a pair of centrioles.

Cutaway diagram of two centrioles at right angles to each other. Structure is a star-shaped tube made up of 9 sets of triplet sets of microtubules separated by non-tubulin proteins.

Included beneath the diagram is an image of two centrioles positioned at right angles to each other in a cell, so that the (columnar) longitudinal section of one centriole is visible, and a star-shaped cross section of the other appears (diameter = about 0.25 micrometers). Networks of single microtubules are visible in the background in the cell.

Most animal cells have a centrosome, a region near the nucleus where the cell's microtubules are initiated. Within the centrosome is a pair of centrioles, each about 250 nm (0.25 μm) in diameter. The two centrioles are at right angles to each other, and each is made up of nine sets of three microtubules. The blue portions of the drawing represent nontubulin proteins that connect the microtubule triplets.

VISUAL SKILLS *How many microtubules are in a centrosome? In the drawing, circle and label one microtubule and describe its structure. Circle and label a triplet.*

*For suggested answer, see*Appendix A.**

Cilia and Flagella

Some eukaryotic cells have **flagella** (singular, *flagellum*) and **cilia** (singular, *cilium*), cellular extensions that contain microtubules. A specialized arrangement of the microtubules is responsible for the beating of these structures. (The bacterial flagellum, shown in Figure 6.5, has a completely different structure.) Many unicellular protists are propelled through water by cilia or flagella that act as locomotor appendages, and the sperm of animals, algae, and some plants have flagella. When cilia or flagella extend from cells that are attached tightly together in a sheet that is part of a tissue layer, they can move fluid over the surface of the tissue. For example, the ciliated lining of the trachea (windpipe) sweeps mucus containing trapped debris out of the lungs (see the EMs in Figure 6.3). In a woman's reproductive tract, the cilia lining the oviducts help move an egg toward the uterus.

Motile cilia usually occur in large numbers on the cell surface. Flagella are usually limited to just one or a few per cell, and they are longer than cilia. Flagella and cilia differ in their beating patterns. A flagellum has an undulating motion like the tail of a fish. In contrast, cilia have alternating power and recovery strokes, much like the oars of a racing crew boat **(Figure 6.23)**.

Figure 6.23

A comparison of the beating of flagella and motile cilia.

A comparison of the beating of flagella and motile cilia.

Part a. Motion of flagella. A flagellum usually undulates, its snakelike motion driving a cell in the same direction as the axis of the flagellum. Propulsion of a human sperm cell is an example of flagellate locomotion (LM).

Part b. Motion of cilia. Cilia have a back-and-forth motion. The rapid power stroke moves the cell in a direction perpendicular to the axis of the cilium. Then, during the slower recovery stroke, the cilium bends and sweeps sideways, closer to the cell surface. A dense nap of cilia, beating at a rate of about 40 to 60 strokes a second, covers this Colpidium, a freshwater protist (colorized SEM).

Video

Flagellum Beating with ATP

image

Play

image

Video

Paramecium Cilia

image

Play

image

A cilium may also act as a signal-receiving “antenna” for the cell. Cilia that have this function are generally nonmotile, and there is only one per cell. (In fact, in vertebrate animals, it appears that almost all cells have such a cilium, which is called a *primary cilium*.) Membrane proteins on this kind of cilium transmit molecular signals from the cell’s environment to its interior, triggering signaling pathways that may lead to changes in the cell’s activities. Cilium-based signaling appears to be crucial to brain function and to embryonic development.

Though different in length, number per cell, and beating pattern, motile cilia and flagella share a common structure. Each motile cilium or flagellum has a group of microtubules sheathed in an extension of the plasma membrane **(Figure 6.24a)**. Nine doublets of microtubules are arranged in a ring with two single microtubules in its center **(Figure 6.24b)**. This arrangement, referred to as the “9+2” pattern, is found in nearly all eukaryotic flagella and motile cilia. (Nonmotile primary cilia have a “9+0” pattern, lacking the central pair of microtubules.) The microtubule assembly of a cilium or flagellum is anchored in the cell by a **basal body**, which is structurally very similar to a centriole, with microtubule triplets in a “9+0” pattern

(Figure 6.24c). In fact, in many animals (including humans), the basal body of the fertilizing sperm's flagellum enters the egg and becomes a centriole.

Figure 6.24

```
Structure of a flagellum or motile cilium.
```

Structure of a flagellum or motile cilium.

Part a. Microtubule doublets appear as circle of 9 double rings or figure 8s, with two single rings in the center of the circle (TEM). Radius of ring = about 0.1 micrometers. Animal cell with cilia; arrows flow to TEM image of cilium. Microtubules labeled inside the plasma membrane. Basal body labeled at base; arrows zoom in on this structure in c.

Part b. Microtubule doublets appear as circle of 9 double rings or figure eights, with two single rings in the center of the circle (TEM). Radius of ring = about 0.1 micrometers. A cross section through a motile cilium shows the 9 + 2 arrangement of microtubules (TEM). The outer microtubule doublets are held together with the two central microtubules by flexible cross-linking proteins (blue in art), including the radial spokes. The doublets also have attached motor proteins called dyneins (red in art). Structures labeled on both the TEM image and the diagram of the cross section through the motile cilium as follows: Plasma membrane wraps outside the cilium. Outer microtubule doublets form a ring around the two central microtubules and are held to the central microtubules by radial spokes. Doublets also have attached motor proteins which are Y shaped in the diagram.

Part c. Cross-section of a basal body: 9 triplets (three joined rings) are joined by non-tubulin proteins to form a star-shaped ring. Proteins form spokes from each triplet as in a wheel. Basal body: The nine outer doublets of a cilium or flagellum extend into the basal body, where each doublet joins another microtubule to form a ring of nine triplets. Each triplet is connected to the next by non-tubulin proteins (thinner blue lines in diagram). This is a 9 + 0 arrangement: The two central microtubules are not present because they terminate above the basal body (TEM).

DRAW IT *In (a) and (b), circle and label the central pair of microtubules. In (a), show where they terminate, and explain why they aren't seen in the cross section of the basal body in (c).*

*For suggested answer, see*Appendix A.**

Animation

Cilia and Flagella

image

Play

image

Play

How does the microtubule assembly produce the bending movements of flagella and motile cilia? Bending involves large motor proteins called **dyneins** (red in the diagram in Figure 6.24) that are attached along each outer microtubule doublet. A typical dynein protein has two “feet” that “walk” along the microtubule of the adjacent doublet, using ATP for energy. One foot maintains contact, while the other releases and reattaches one step farther along the microtubule (see Figure 6.21). The outer doublets and two central microtubules are held together by flexible cross-linking proteins (blue in the diagram in Figure 6.24), and the walking movement is coordinated so that it happens on one side of the circle at a time. If the doublets were not held in place, the walking action would make them slide past each other. Instead, the movements of the dynein feet cause the microtubules—and the organelle as a whole—to bend.

Microfilaments (Actin Filaments)

Microfilaments are thin solid rods. They are also called actin filaments because they are built from molecules of **actin**, a globular protein. A microfilament is a twisted double chain of actin subunits (see Table 6.1). Besides occurring as linear filaments, microfilaments can form structural networks when certain proteins bind along the side of such a filament and allow a new filament to extend as a branch. Like microtubules, microfilaments seem to be present in all eukaryotic cells.

In contrast to the compression-resisting role of microtubules, the structural role of microfilaments in the cytoskeleton is to bear tension (pulling forces). A three-dimensional network formed by microfilaments just to the inside of the plasma membrane (*cortical microfilaments*) helps support the cell’s shape (see Figure 6.8). This network gives the outer cytoplasmic layer of a cell, called the **cortex**, the semisolid consistency of a gel, in contrast with the more fluid state of the interior cytoplasm. In some kinds of animal cells, such as nutrient-absorbing intestinal cells, bundles of microfilaments make up the core of microvilli, delicate projections that increase the cell’s surface area **(Figure 6.25)**.

Figure 6.25

```
A structural role of microfilaments.
```

On a 0.25 nanometer scale, microvilli extend from intermediate filaments, reinforced by microfilaments (actin filaments) and with plasma membranes.

The surface area of this nutrient-absorbing intestinal cell is increased by its many microvilli (singular, microvillus), cellular extensions reinforced by bundles of microfilaments. These actin filaments are anchored to a network of intermediate filaments (TEM).

Microfilaments are well known for their role in cell motility. Thousands of actin filaments and thicker filaments made of a protein called **myosin** interact to cause contraction of muscle cells **(Figure 6.26a)**; muscle contraction is described in detail in Concept 50.5. In the unicellular protist *Amoeba* and some of our white blood cells, localized contractions brought about by actin and myosin are involved in the amoeboid (crawling) movement of the cells. The cell crawls along a surface by extending cellular extensions called **pseudopodia** (from the Greek *pseudes*, false, and *pod*, foot) and moving toward them **(Figure 6.26b)**. In plant cells, actin-protein interactions contribute to **cytoplasmic streaming**, a circular flow of cytoplasm within cells **(Figure 6.26c)**. This movement, which is especially common in large plant cells, speeds the movement of organelles and the distribution of materials within the cell.

Figure 6.26

```
Microfilaments and motility.
```

Microfilaments and motility.

In these three examples, interactions between actin filaments and motor proteins bring about cell movement.

Part a. Myosin motors in muscle cell contraction. The walking of myosin projections (the so-called heads) drives the parallel myosin and actin filaments past each other so that the actin filaments approach each other in the middle (red arrows). This shortens the muscle cell. Muscle contraction involves the shortening of many muscle cells at the same time (TEM).

Part b. Amoeboid movement. Interaction of actin filaments with myosin causes contraction of the cell, pulling the cell's trailing end (at left) forward (to the right) (LM). Amoeboid cell (diameter: about 500 micrometers) with labels: inner cytoplasm, cortex (outer cytoplasm) gel within actin network, and extending pseudopodium, or projections. Arrows indicate direction of movement toward the right.

Part c. Cytoplasmic streaming in plant cells. A layer of cytoplasm cycles around the cell, moving over tracks of actin filaments. Myosin motors attached to some organelles drive the streaming by interacting with the actin (LM). Plant cells (diameter: about 60 micrometers) with cytoplasm and chloroplasts visible. Arrows indicate the direction of streaming within the cells.

BioFlix® Animation

Actin and Myosin in Muscle Contraction

image

image

Video

Amoeba Pseudopodia

image

Play

image

Video

Cytoplasmic Streaming

image

Play

image

Intermediate Filaments

Intermediate filaments are named for their diameter, which is larger than the diameter of microfilaments but smaller than that of microtubules (see Table 6.1). While microtubules and microfilaments are found in all eukaryotic cells, intermediate filaments are only found in the cells of some animals, including vertebrates. Specialized for bearing tension (like microfilaments), intermediate filaments are a diverse class of cytoskeletal elements. Each type is constructed from a particular molecular subunit belonging to a family of proteins whose members include the keratins. Microtubules and microfilaments, in contrast, are consistent in diameter and composition in all eukaryotic cells.

Intermediate filaments are more permanent fixtures of cells than are microfilaments and microtubules, which are often disassembled and reassembled in various parts of a cell. Even after cells die, intermediate filament networks often persist; for example, the outer layer of our skin consists of dead skin cells full of keratin filaments. Intermediate filaments are especially sturdy and play an important role in reinforcing the shape of a cell and fixing the position of certain organelles. For instance, the nucleus typically sits within a cage made of intermediate filaments. Other intermediate filaments make up the nuclear lamina, which lines the interior of the nuclear envelope (see Figure 6.9). In general, the various kinds of intermediate filaments seem to function together as the permanent framework of the entire cell.